Bill of Materials

Duotao Gao · EGR 304 · Team 103 GripRx · User Interface Board

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Team website | Individual home

Overview

This bill of materials covers the UI subsystem: PIC18F57Q43 Curiosity Nano, 5 V supply, LCD, user controls, MCP6004 audio circuit, speaker, connectors and debugging hardware. Provided materials remain listed. Purchases and Peralta stock requests are separated to avoid duplicate orders.

Budget summary

Measure	Amount / treatment
Digi-Key order, including planned spares	$37.59
Digi-Key parts for one UI subsystem, without spares	$14.81
Peralta materials	Excluded from the course $60 per-member allowance
Development kits, free parts and out-of-pocket purchases	Excluded by course policy
Tax and shipping	Not quoted; budget treatment to be confirmed by instructor

The $37.59 order is below the $60 per-member course allowance. It leaves $22.41 before any charges that the instructor determines should count. The $14.81 figure is the purchased-parts contribution for one UI subsystem, excluding Peralta and supplied inventory; it is not a valuation of every material. The team MF-05 target is $150 for the prototype and must be checked using the combined team BOM.

Digi-Key purchase request

Refs	Part / description	Manufacturer / MPN	Order qty	Unit USD	Total USD
J4	16 x 2 character LCD module	Newhaven Display / NHD-0216K1Z-FSW-FBW-L	2	$8.11	$16.22
C14	Electrolytic capacitor, 220 uF 16 V	Rubycon / 16YXJ220MCR6.3X11	4	$0.45	$1.80
C2	Nonpolar ceramic capacitor, 0.33 uF 50 V	TDK / FA28X7R1H334KRU00	4	$0.37	$1.48
C9	Nonpolar ceramic capacitor, 1 uF 50 V (16 V minimum)	TDK / FG28X5R1H105KRT00	4	$0.31	$1.24

Refs	Part / description	Manufacturer / MPN	Order qty	Unit USD	Total USD
R13	Resistor, 200k	YAGEO / MFR-25FBF52-200K	4	$0.11	$0.44
D3	Schottky diode, reverse-polarity protection	Taiwan Semiconductor / 1N5822	4	$0.71	$2.84
J1	Shrouded ribbon header, 2 x 4	Wurth Elektronik / 61200821621	3	$1.03	$3.09
J5	Speaker screw terminal, 2 positions	On Shore Technology / OSTTE020161	3	$0.67	$2.01
External speaker via J5	Speaker, 8 ohm, 2 W	Same Sky / CMS-3652-28L200	3	$2.17	$6.51
U1 heatsink	TO-220 heatsink	Advanced Thermal Solutions / ATS-PCBT1085	2	$0.68	$1.36
JP1, JP2 shunts	Two-pin jumper shunt	Sullins / SPC02SYAN	6	$0.10	$0.60

Digi-Key materials subtotal: $37.59.

Peralta stock request

The remaining seven Peralta requests follow the course-designated stock planning sheet. Quantities below are requests, not confirmation of receipt. The listed row numbers identify entries in the Fall 2022 worksheet; they are not physical shelf/bin numbers. Unit charges are not quoted, and no retail replacement price is assigned to unmarked stock.

Refs	Required material	Requested qty	Source row
F1	1.5 A, 5 x 20 mm cartridge fuse	4	87
J3	PJ-102AH through-hole DC barrel jack	3	64
RV1, RV2	10 kohm potentiometer, top-up after two owned units	4	54
U3 socket	14-pin DIP socket, 7.62 mm row spacing	3	78
J4 PCB socket	40-position 2.54 mm female header strip	2	76
F1 holder	5 x 20 mm PCB fuse holder	3	63
Shared male header stock	40-position 2.54 mm breakaway male header strip	3	75

Complete component list

The list includes every schematic component and the required supplementary hardware. “Available” means reported available by the student; kit manufacturers and exact spare counts were not supplied. “Class-issued” follows the provided assignment planning reference.

Refs	Part / description	Qty / system	Request qty	MPN / specification	Source / status
C3	Nonpolar capacitor, 0.1uF 50V	1	0	—	Already available; no new order
C5, C8, C16, C17, C18, C19	Nonpolar capacitor, 0.1uF 16V	6	0	—	Already available; no new order
C15	Nonpolar capacitor, 0.1uF 16V(LCD VDD)	1	0	—	Already available; no new order
D1	GreenLED Power)	1	0	—	Already available; no new order
D2	Red LED(debug)	1	0	—	Already available; no new order
R1, R11	Resistor, 1k 1/4W	2	0	—	Already available; no new order
R5	Resistor, 100k 1/4W	1	0	—	Already available; no new order
R7, R8, R9, R10	Resistor, 10k 1/4W	4	0	—	Already available; no new order
R12	Resistor, 330 1/4W	1	0	—	Already available; no new order
R14, R15	Resistor, 10k	2	0	—	Already available; no new order
R16	Resistor, 330R / 0.25w	1	0	—	Already available; no new order
SW1	Momentary pushbutton, START/STOP	1	0	—	Already available; no new order
SW2	Momentary pushbutton, UP	1	0	—	Already available; no new order
SW3	Momentary pushbutton, DOWN	1	0	—	Already available; no new order
SW4	Momentary pushbutton, DEBUG	1	0	—	Already available; no new order
U1	5 V linear voltage regulator	1	0	L7805ABV-DG	Already available; no new order
U3	Quad operational amplifier, PDIP-14	1	0	MCP6004-I/P	Already available; no new order
U2 mating headers	Female headers for Curiosity Nano	2	0	—	Already available; no new order
External power adapter	External wall adapter, 9 V, 3 A	1	0	B09ZTKTLGW	Class-issued stock (confirm quantity)
U2	PIC18F57Q43 Curiosity Nano board	1	0	DM164150	Confirm class-issued stock
C2	Nonpolar ceramic capacitor, 0.33 uF 50 V	1	4	FA28X7R1H334KRU00	Digi-Key
C9	Nonpolar ceramic capacitor, 1 uF 50 V (16 V minimum)	1	4	FG28X5R1H105KRT00	Digi-Key
C14	Electrolytic capacitor, 220 uF 16 V	1	4	16YXJ220MCR6.3X11	Digi-Key
D3	Schottky diode, reverse-polarity protection	1	4	1N5822	Digi-Key
J1	Shrouded ribbon header, 2 x 4	1	3	61200821621	Digi-Key
J4	16 x 2 character LCD module	1	2	NHD-0216K1Z-FSW-FBW-L	Digi-Key

Refs	Part / description	Qty / system	Request qty	MPN / specification	Source / status
J5	Speaker screw terminal, 2 positions	1	3	OSTTE020161	Digi-Key
R13	Resistor, 200k	1	4	MFR-25FBF52-200K	Digi-Key
External speaker via J5	Speaker, 8 ohm, 2 W	1	3	CMS-3652-28L200	Digi-Key
U1 heatsink	TO-220 heatsink	1	2	ATS-PCBT1085	Digi-Key
JP1, JP2 shunts	Two-pin jumper shunt	2	6	SPC02SYAN	Digi-Key
J6	Spare GPIO header1x10, 2.54 mm	1	0	Cut from Peralta male strips	Included in shared strip allocation
JP1	Two-pin jumper header, +5V-VTG	1	0	Cut from Peralta male strips	Included in shared strip allocation
JP2	Two-pin jumper header, VOFF-GND	1	0	Cut from Peralta male strips	Included in shared strip allocation
TP1	Test point	1	0	Cut from Peralta male strips	Included in shared strip allocation
TP2, TP3, TP4, TP5, TP6	Test point	5	0	Cut from Peralta male strips	Included in shared strip allocation
J4 LCD male header	LCD male pin header, 1 x 16	1	0	Cut from Peralta male strips	Included in shared strip allocation
F1	1.5 A, 5 x 20 mm cartridge fuse	1	4	Course-list specification	Peralta request
J3	PJ-102AH through-hole DC barrel jack	1	3	Course-list specification	Peralta request
RV1, RV2	10 kohm potentiometer, top-up after two owned units	2	4	Course-list specification	Peralta request
U3 socket	14-pin DIP socket, 7.62 mm row spacing	1	3	Course-list specification	Peralta request
J4 PCB socket	LCD 1x16 female socket, cut from 40-position stock	1	2 x 40-position strips	Course-list specification	Peralta request
F1 holder	5 x 20 mm PCB fuse holder	1	3	Course-list specification	Peralta request
Shared male header stock	40-position 2.54 mm breakaway male header strip	Procurement stock	3	2.54 mm breakaway male strips	Peralta request
J1 cable	Ribbon cable assembly, 8 conductors, 2 x 4 IDC ends	1	Team coordination	Team selection pending	Shared cable; purchaser/length not yet established

Quantity and package notes

- Quantities cover breadboard testing, PCB assembly and the planned spare allowance. Already-owned U1, U3, matching Nano headers and kit parts are not repurchased. Two 10 kohm pots are owned; four additional pots are requested to reach six.
- Two LCDs are requested: one can move from the breadboard to a socketed PCB, while the second provides a spare.
- Three 40-pin male strips provide 120 pins: LCD headers 48, J6 headers 30, jumper headers 12 and test points 14. Total allocated 104, leaving 16 spare. These pins are counted once.
- Two 40-position female strips provide three 16-position LCD sockets; allow sacrificial contacts when cutting.

- C14 is the selected Rubycon 220 uF 16 V part, D6.3 mm with 2.5 mm pitch; verify positive pad and 12.5 mm maximum seated-height clearance. C2 is the selected TDK 0.33 uF 50 V part, 5 mm pitch. Match the actual issued potentiometer and fuse-holder packages.
- Verify fuse response, DC plug dimensions/polarity, connector keying and all pin mappings. LCD module approval and the team cable are instructor/team review items.
- C9 remains a purchased nonpolar 1 uF ceramic. Polarized tantalum stock is not substituted. The specified 1N5822 and 200 kohm resistor remain on the purchase request.

Manufacturer documents and purchasing links

Refs	MPN	Manufacturer document	Product / source
C2	FA28X7R1H334KRU00	PDF	Product page
C9	FG28X5R1H105KRT00	PDF	Product page
C14	16YXJ220MCR6.3X11	PDF	Product page
D3	1N5822	PDF	Product page
J1	61200821621	PDF	Product page
J4	NHD-0216K1Z-FSW-FBW-L	PDF	Product page
J5	OSTTE020161	PDF	Product page
R13	MFR-25FBF52-200K	PDF	Product page
U1	L7805ABV-DG	PDF	Product page
U2	DM164150	PDF	Product page
U3	MCP6004-I/P	PDF	Product page
External speaker via J5	CMS-3652-28L200	PDF	Product page
U1 heatsink	ATS-PCBT1085	PDF	Product page
JP1, JP2 shunts	SPC02SYAN	PDF	Product page

Provided generic parts use the specifications in the course stock list. Manufacturer family documents are references; select PCB footprints from the actual issued part. Archived alternative documents do not establish the identity of unmarked stock.

Alternate candidates

Use	Candidate	Review needed
LCD	NHD-0216K1Z-FLYBW	Voltage, pinout, dimensions and course approval
U1	LM2940CT-5.0	Different capacitor/stability requirements

Use	Candidate	Review needed
U3	MCP6024-I/P	Operating range, output performance and pinout
Speaker	CMS-4065-28L200	Dimensions, impedance and sound output
External adapter	GST18A09-P1J	Plug dimensions, polarity and course approval

These are candidates for review, not pre-approved interchangeable parts.

Files

- Full BOM spreadsheet
- BOM page PDF
- Digi-Key request XLSX · PDF
- Peralta request XLSX · PDF
- Digi-Key bulk-upload CSV

AI assistance

ChatGPT assisted with transcription of supplied schematic/BOM information, price lookup, quantities, source classification, calculations and document formatting. The student supplied inventory information and the email address. Actual part receipt, physical compatibility and instructor approval are not claimed. Prompts and disclosure.

Resources

- Individual Hardware Orders
- Course Sequence Requirements